

# OU accumulator fits to the gain-modulated RNNs, round 2: a control that cannot see the transition and a fit that does

Ivan Grahek, 24 September 2026 (v2, with per-coherence posterior predictives). Code, tables and RESULTS.md notes: branches track-a-hssm-ou-corrected, track-b-evidence-conditioned-fit and round2-report of rnn\_hssm (issues #3 and #4). HSSM 0.2.4, ssm-simulators 0.8.3, gain-controller-rnn at 1ca084b.

## 1. The question

A scalar gain on the transfer slope of the trained NXX1 units moves the network between integration regimes: a leaky integrator at gain 0.8, close to a line attractor at 1.0, and a repelling undecided state with two committed wells at 1.2. The landscape analysis of 21 September measures the Langevin drift of the choice variable  $dv$  on noise-only trials; its slope at the undecided state is about -4.3, +1.6 and +6.2 per second at gains 0.8, 1.0 and 1.2 (three networks). Behaviourally, the psychophysical kernel of Figure 1F of the manuscript sweeps from recency to primacy across gain, with its slope crossing zero at gain 1.016 over twenty networks.

The accumulator version of the claim is an Ornstein-Uhlenbeck (OU) process,  $dx = (v - g \cdot x) dt + dW$ , whose leak  $g$  goes from positive through zero to negative as gain rises. Throughout,  $g$  follows the convention of ssm-simulators, the simulator HSSM's OU likelihood was trained on, whose update is  $x += (v - g \cdot x) \cdot dt + \text{noise}$  (src/cssm.pyx, line 1720 at tag 0.8.3): positive  $g$  is leaky and gives recency, negative  $g$  is unstable and gives primacy. This is the negative of Brunton's  $\lambda$  and of the landscape drift slope. Stretching RTs by  $k$  describes the same process with  $g/k$ ,  $av/k$  and  $v/v/k$ . Both were re-verified by simulation for this report (median first-passage times 0.59, 0.78 and 1.10 s at  $g = -1, 0, +1$ ; stretched RT quantiles within 4 percent of native at  $k = 10$ ).

Round 2 asks two questions. Once every defect of the original HSSM fits is removed, does a fit to RT and choice distributions see the transition? And does a likelihood fit of the same accumulator, conditioned on each trial's evidence stream, see it per network with uncertainty? The first is the control for the second.

## 2. The original fits and why they failed

The original three fits (one network, pretrained HSSM OU likelihood) diverged on nearly every draw, with  $g$  at the unstable edge of its range, drift at the upper edge and the fitted non-decision time absorbing the 0.3 s offset. Four problems were separated. The data were read out through a Weibull collapsing bound, which a constant-bound OU can only imitate with unstable dynamics; RT and choice were therefore regenerated from the same  $dv$  traces with a constant bound of 1.5, at the cost of 3.4, 0.3 and 0.0 percent omitted trials at the three gains. The networks decide in 75 to 145 ms with a leak of several per second, outside the likelihood's box of  $|g|$  at most 1 per second and decision times near a second, so a time stretch of 8 to 10 is needed and none was used. Non-decision time was fitted. And the 750 ms horizon truncates the data, which the recovery study showed pushes  $g$  toward unstable.

## 3. The control: corrected HSSM fits to RT and choice

Track A refits the constant-bound data (bound 1.5 on  $dv$ , twenty networks, 38 640 to 39 988 trials per gain) with HSSM's pretrained OU likelihood and every defect removed. All RTs are multiplied by  $k = 10$  and offset by 0.3 s. Non-decision time is fixed, not fitted, at the offset plus the minimum RT less one native millisecond so the fastest trial keeps a positive decision time. The lapse mixture is off. Drift varies with coherence through  $v \sim 1 + \text{signed coherence}$  with an identity link, so drift  $= v_0 + v_1 \cdot \text{coherence}$ , and the bound  $a$ , start point  $z$  and leak  $g$  are shared across coherences within a gain. Choice is coded as the network's choice and priors are uniform over the likelihood's box. Three pooled fits at each of  $k = 8, 10, 12, 16, 20$  and 24, 120 per-network fits, three hierarchical fits with  $v$ ,  $a$  and  $g$  varying by network, six recovery fits and two appendix fits were run; every one has R-hat at most 1.01 and bulk effective sample size above 400, with one divergence in 600 000 draws.

The sign is unambiguous and the magnitude is not. At gains 0.8 and 1.0 the posterior for  $g$  sits on the +1 edge of the likelihood at every stretch from 8 to 24, so its native value equals  $k$  and is a floor, not an estimate. Over the same three-fold range of  $k$  the native bound and drift are stable to about ten percent, as the rescaling rule requires of identified parameters. At gain 1.2 the posterior leaves the edge from  $k = 10$  on and its native value rises from +9.2 to +14.7 per s between  $k = 10$  and 24. The hierarchical fits converge and agree with the pooled ones within 2 percent.

| gain | g, stretched [94% HDI]  | g native (per s) | a native | drift native at coherence 0.15 | posterior mass at the +1 ceiling | networks with g > 0 |
|------|-------------------------|------------------|----------|--------------------------------|----------------------------------|---------------------|
| 0.8  | +0.998 [+0.994, +1.000] | +10.0            | 0.38     | 3.8                            | 1.00                             | 20 of 20            |
| 1.0  | +0.996 [+0.988, +1.000] | +10.0            | 0.32     | 4.1                            | 0.99                             | 20 of 20            |
| 1.2  | +0.921 [+0.872, +0.968] | +9.2             | 0.29     | 4.1                            | 0.01                             | 20 of 20            |

Table 1. Pooled HSSM fits at  $k = 10$  (output/track\_a/headline\_pooled.csv). Positive  $g$  is leaky; native units follow  $g \cdot k$ ,  $a/\sqrt{k}$  and  $v/\sqrt{k}$ . The last column counts the 120 per-network fits at the same stretch; 118 have a 94 percent HDI excluding zero on the leaky side.

Why the leak runs to the ceiling is visible in the data. The networks were trained on a fixed-duration task and read out at 750 ms; they do not produce RTs. Every RT here is imposed afterwards by a threshold on the dv trace, and dv has two timescales: the slow choice mode whose leak or instability gain changes, and fast mean-reverting fluctuations from the readout units. A threshold inside the noise band is crossed early and often by the fast component, so the RT distribution has a leading edge of a few milliseconds and a flat tail to the horizon, and strong mean reversion is the only way a one-leak OU produces both shapes at once. Raising the threshold so that the slow mode governs the crossing removes most low-gain trials as omissions and the single-timescale OU stops fitting what remains (Table 2). The fitted leak moves by an order of magnitude with that arbitrary readout choice, which is the clearest sign that it is not measuring the network's integration.

| constant bound on dv                 | median RT, gains 0.8 / 1.0 / 1.2 (ms) | omissions by 750 ms | exact-simulator g at gain 1.2 (per s) | fit loss at gain 1.2 |
|--------------------------------------|---------------------------------------|---------------------|---------------------------------------|----------------------|
| 1.5 (used in Table 1)                | 145 / 99 / 76                         | 3.4 / 0.3 / 0.0 %   | +9.3                                  | 0.14                 |
| 2.0                                  | 203 / 136 / 102                       | 13 / 1.7 / 0.2 %    | +8.6                                  | 0.62                 |
| 2.5                                  | 255 / 171 / 127                       | 31 / 10 / 5.6 %     | +5.2                                  | 2.2                  |
| 2.94 (the Weibull's starting height) | 321 / 229 / 166                       | 67 / 30 / 14 %      | +0.5                                  | 10.3                 |

Table 2. The RT observable depends on the threshold placed on dv (output/fixed\_bound/sanity.csv, output/feasibility/matches\_table.csv). The last two columns are the exact-simulator OU fit to RT and choice at gain 1.2 with no parameter box; a loss below 0.5 is indistinguishable from the data by eye. The original Weibull readout starts at 2.94 and collapses, giving task-like RTs (medians 373 / 258 / 183 ms) by forcing late choices toward chance.

The model reproduces what it was fitted to (Figure 1). Simulating 20 000 trials per coherence from 200 posterior draws gives median absolute RT quantile errors of 6.5, 1.6 and 0.8 ms on correct trials at gains 0.8, 1.0 and 1.2, accuracy within 0.02 at every coherence and omissions within 1.6 points. The one visible misfit is the error trials at high coherence for gain 0.8, which the network makes as fast as its correct responses and a leaky OU makes slower. The posterior predictive used an Euler integrator of the same process because the ssm-simulators C code corrupts its heap when called repeatedly with changing parameters; on a cross-check the two engines agree within Monte Carlo noise.

The decisive check drives the fitted model with the networks' own evidence streams and computes the psychophysical kernel (Figure 2). With its fitted bound, 0.29 to 0.38 in native units and reached within about 100 ms on more than 99.9 percent of trials, the model commits before most of the evidence arrives and shows primacy at every gain, slopes -0.041, -0.057 and -0.064 against the networks' +0.039, +0.002 and -0.026; what ordering it has follows the falling bound. With the bound removed the fitted leak gives recency at every gain, slopes +0.068, +0.067 and +0.066, because the fitted  $g$  is the same large number at all three gains. Neither variant reproduces any gain or the ordering. These slopes were reproduced independently for this report to within 0.003.

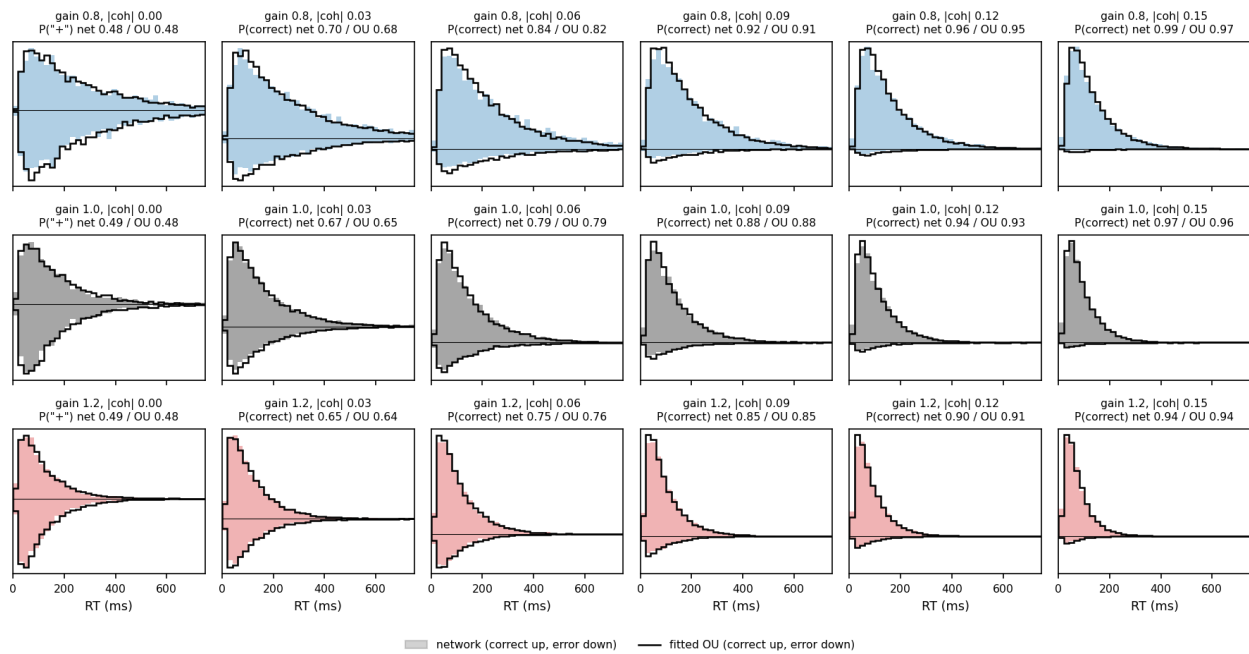


Figure 1. Track A posterior predictive on RT and choice, by [coherence] (columns) and gain (rows). Network RT histograms (filled) against the HSSM-fitted OU at its pooled posterior mean (line), twenty networks pooled, constant bound 1.5. Correct trials are plotted upward and errors downward, and within each panel the two sum to one for the network and for the model, so the areas give the accuracy. Titles give accuracy for network and model; at coherence 0 the split is by choice.

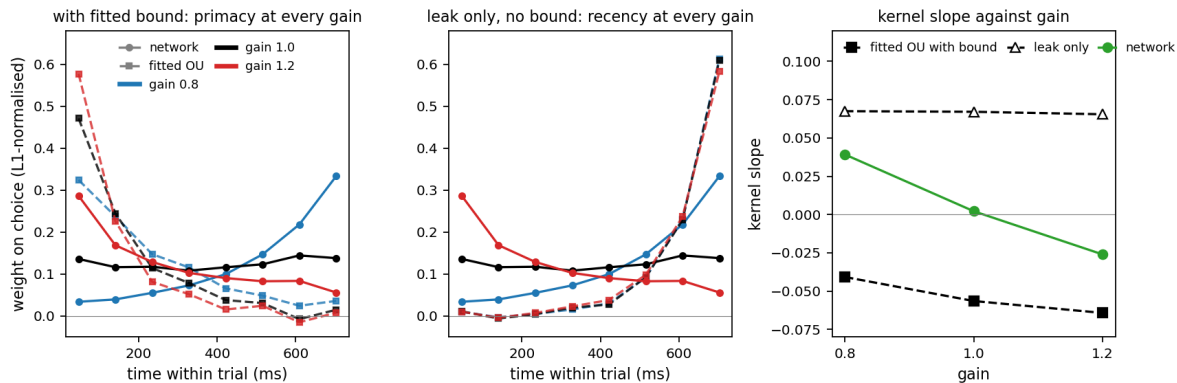


Figure 2. Track A posterior predictive on the psychophysical kernel. Networks (solid) against the fitted OU driven by the same evidence streams (dashed), with its fitted bound (left) and without (middle); kernel slope against gain for both variants (right). Positive g is leaky.

Two checks bound the reading. The pipeline is not blind to instability: with data simulated from the fitted parameters and g replaced by the theoretical values, the refit recovers the sign in all six datasets and at gain 1.2 the HDI excludes zero on the unstable side; magnitudes shrink toward zero by about a third, so only one of six HDIs covers the truth and Table 1 should be read as signs. And the original sign flip is a confound: the same network at the same gain fitted on the Weibull-bound readout gives  $g = -0.22 [-0.28, -0.15]$  at  $k = 10$ .

#### 4. The result: the same accumulator fitted to choice given the evidence stream

Track B fits the same OU in the network's own time ( $dt = 1$  ms) to each trial's deadline choice given that trial's evidence sequence, after Brunton, Botvinick and Brody (2013):  $a_{t+1} = a_t + (v e_t - g a_t) dt + \sqrt{dt} \xi_t$ ,  $a_0 = a_{\text{bias}}$ , an optional sticky bound at  $\pm B$ , choice =  $\text{sign}(a_T)$ , noise scale fixed at one so  $v$  and  $B$  are in noise units. Without the bound  $a_T$  is Gaussian given the evidence, the likelihood of each choice is a normal probability and a fit takes seconds. It was fitted by maximum likelihood and by NUTS with weak priors for every network and gain, and pooled over networks; the analytic likelihood matches Monte Carlo within 0.005 and an independent implementation within 0.003 in  $g$ .

| gain | median g across networks (per s) | interquartile range | intervals excluding zero on the predicted side | pooled g [94% HDI]   | landscape coefficient |
|------|----------------------------------|---------------------|------------------------------------------------|----------------------|-----------------------|
| 0.8  | +4.13                            | +3.02 to +4.63      | 20 of 20 positive                              | +3.95 [+3.83, +4.06] | about +4.3            |
| 1.0  | +0.32                            | -0.34 to +0.79      | 10 positive, 5 negative                        | +0.23 [+0.15, +0.32] | about -1.6            |
| 1.2  | -2.63                            | -3.56 to -2.03      | 20 of 20 negative                              | -2.73 [-2.83, -2.62] | about -6.2            |

Table 3. Track B, unbounded evidence-conditioned fit (output/track\_b/analytic\_per\_network.csv, analytic\_pooled.csv). Positive g is leaky. Intervals are 94 percent HDIs from NUTS; maximum-likelihood estimates agree in sign in all 60 cells. The landscape coefficient is the sign-converted drift slope from three networks.

The sign pattern is the predicted one in every network at the extreme gains: twenty of twenty leaky at gain 0.8 and twenty of twenty unstable at gain 1.2, with intervals excluding zero in every case. At gain 1.0 the networks straddle zero with a slightly positive pooled estimate, and the zero crossing of g against gain falls at 1.016 pooled (per-network median 1.023), where Figure 1F puts the kernel's. The per-network g follows each network's own kernel slope with  $r = 0.996$  (Figure 3, right). The pooled values match the first session's simulation-matching estimates (+3.98, +0.24, -2.50) and the ordering and zero crossing of the landscape, with a smaller magnitude at gain 1.2 because the network saturates into wells near  $|dv| = 3$  that a linear process lacks.

The fitted model reproduces the kernel it was not fitted to: pooled slopes +0.041, +0.002 and -0.031 against the networks' +0.039, +0.002 and -0.026, per-network slopes on the identity line, psychometric curves overlapping (Figure 4). The one shortfall is at gain 1.2, where the network's kernel falls faster over the first 200 ms than a linear unstable process can, the mark of early commitment into a well. Read the way an experiment would be, the fit also reproduces the averaged data (Figure 5): the network's  $P(\text{choice})$  at every signed coherence to within 0.015, accuracy by  $|\text{coherence}|$  on the network mean, and calibration of the per-trial predicted probabilities within 0.008 per decile over 40 000 trials per gain. Recovery with the fitted or the theoretical values planted returns the right sign in 60 of 60 refits and covers the truth in 56. Adding the sticky bound and fitting by Monte Carlo likelihood (300 realisations per trial, common random numbers, profile intervals) changes little when the bound is free: medians +4.13, +0.52 and -2.59 per s, 20 of 20 and 19 of 20 on the predicted side, agreement with the unbounded fit network by network at  $r = 0.998, 0.926$  and  $0.952$ . The bound itself is identified only from below (median about 6.4), since choices alone do not say where it is.

## 5. Leak versus bound

A strong leak with early commitment at a low bound gives a primacy kernel just as an unstable accumulator does; three of sixty of the first session's simulation-matching fits landed there with g near +12. With the per-trial likelihood it happens once in sixty: network 61 at gain 1.2 gives  $g = +1.06$  with  $B = 1.5$ , 2.5 log-likelihood units above the unbounded solution at  $g = -0.93$ . A dataset generated from that degenerate solution on real evidence is recovered by the choice-only likelihood ( $g = +11.9 [+10.5, +12.9]$ ), so 2000 trials do separate the regimes and the remaining case is a local optimum.

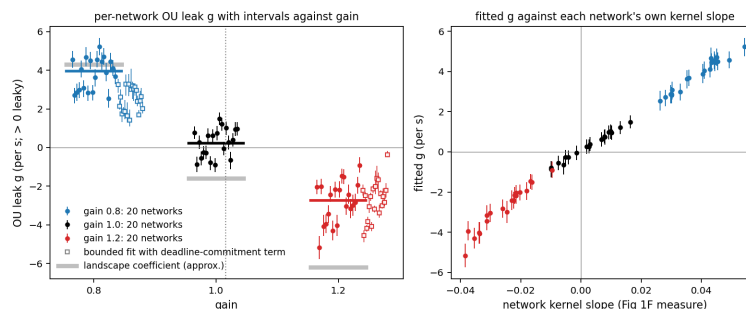


Figure 3. Track B. Left: fitted OU leak g per network with 94 percent intervals against gain (filled, unbounded fit), the bounded fit with the deadline-commitment term (open squares, gains 0.8 and 1.2), pooled fits (coloured bars), landscape coefficients (gray bars) and the kernel zero crossing of Figure 1F at gain 1.016 (dotted). Right: the same estimates against each network's own kernel slope. Positive g is leaky.

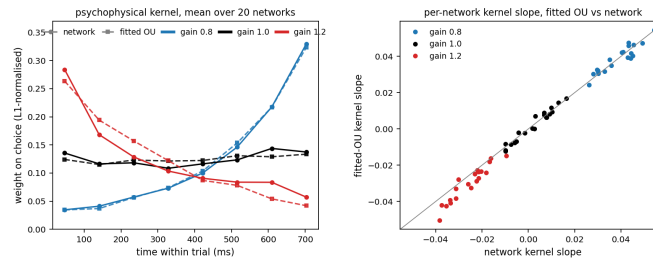


Figure 4. Track B. Left: psychophysical kernels of the networks (solid) and of the fitted OU driven by the same evidence streams (dashed), averaged over the twenty networks. Right: kernel slope of the fitted model against that of the network, one point per network and gain.

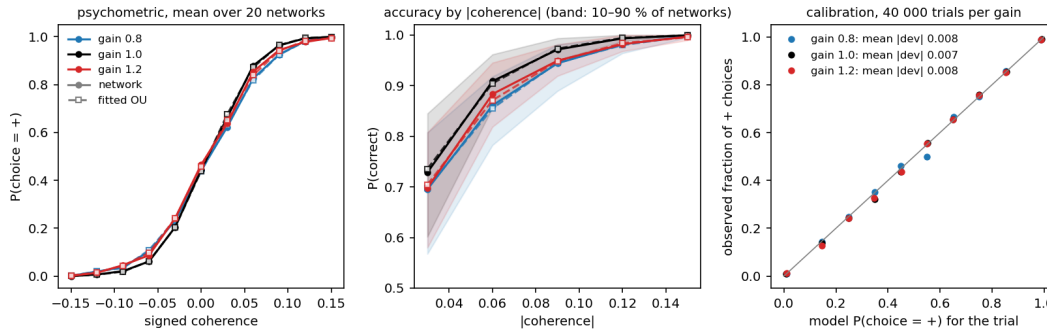


Figure 5. Track B posterior predictive on the averaged data. Left: P(choice = +) by signed coherence, network (filled circles) and fitted OU (open squares), mean over twenty networks. Middle: accuracy by |coherence| with the 10 to 90 percent band across networks. Right: calibration, trials binned by the model's predicted probability against the observed fraction of + choices.

| bounded-fit variant | median g at 0.8 | median g at 1.2 | predicted side, interval excluding 0 | median B at 0.8, 1.2 | pooled kernel slope at 0.8 (network +0.039) |
|---------------------|-----------------|-----------------|--------------------------------------|----------------------|---------------------------------------------|
| no hit term         | +4.13           | -2.59           | 20 of 20, 19 of 20                   | 6.6, 6.4             | +0.040                                      |
| hit at the deadline | +2.51           | -2.53           | 20 of 20, 20 of 20                   | 1.4, 1.8             | +0.021                                      |
| ever crossed 2.0    | +5.90           | -3.51           | 20 of 20, 19 of 20                   | 1.0, 1.4             | not run                                     |
| crossing-time bins  | +8.28           | -1.19           | 20 of 20, 14 of 20                   | 0.9, 0.8             | -0.010                                      |

Table 4. Track B, bounded fits (output/track\_b/route2\_per\_network.csv). Positive g is leaky. The last two rows use the network's transient 2.0 crossing, which a sticky bound cannot represent; they show the failure mode, not estimates.

The issue proposed the fraction of trials whose |dv| reaches 2.0 before the deadline as the observable that breaks the degeneracy. It is the wrong observable for a sticky-bound model, for an informative reason: the network's 2.0 crossing is not absorbing. Of the trials that cross before the deadline, 46, 21 and 10 percent are back below 2.0 at 750 ms at gains 0.8, 1.0 and 1.2. Forcing a sticky bound to match that transient rate drives B below 1, inflates the leak at gain 0.8 (Table 4) and flips the sign of the model's kernel there. Measured at the deadline instead, |dv| above 2.0 at 750 ms, the observable is what a sticky bound implies; that variant identifies B, matches the commitment fraction within 0.015, keeps the kernel, and resolves network 61 to g = -0.36 [-0.50, -0.19].

## 6. What remains

The transition is in the networks' behaviour and an accumulator recovers it, but only when the fit is conditioned on each trial's evidence. What stays model-dependent is the magnitude at gain 0.8, between +2.5 and +4.1 per s depending on how the bound is constrained; the sign and ordering do not move. The leak at gain 1.2 sits near -2.6, below the landscape's -6.2 because a linear OU cannot saturate into the wells. A two-timescale accumulator, a fast mean-reverting readout on a slow choice mode, would reconcile Track A's leak of at least +24 per s with Track B's few per second and is the next model. Not done: a Weibull appendix beyond one network, and Track B on a bounded rather than a deadline choice.